Kestrel and SLM360

An honest comparison: who is behind, in what, and by how much

28 October 2026

The short answer

SLM360 is an on-device language engine made by 360 Labs (Lightyear 360 Labs Pvt Ltd, India). Its main job is to read a short command and pick one action from a fixed list. Kestrel is an offline personal assistant. It also picks actions, but it can write emails, translate, rewrite text, answer questions from your notes and write small programs.

- **SLM360 is far ahead on size and on phones.** Its intent model is about 33 times smaller than Kestrel's (4 MB vs 132 MB), and its text model about 45 times smaller (63 MB vs 2.84 GB). 360 Labs describes Android apps built on it. Kestrel has never run on a phone.
- **Kestrel is ahead on accuracy, languages, abilities and proof.** On the same 150-intent test (CLINC150), Kestrel's intent step scored 95.9% on 4,500 test sentences; SLM360 Nano's page claims 85.1%. SLM360's headline 98–100% comes from tests of only 49 and 50 sentences. Kestrel measured Hinglish; SLM360's paper is English only.
- **Many SLM360 numbers cannot be checked.** The model is not public, no test phone is named, the paper's "raw results" link returns an error (404), and the same model is described with three different sizes.

In one sentence: **SLM360 is small enough for phones but proves little; Kestrel proves what it does but is too big for phones.**

1 How this comparison was made

- **Kestrel:** numbers from the committed logs in this repository, all measured on one laptop: AMD Ryzen 5 5500U, 6 cores, 14 GiB RAM, Fedora 44, CPU only, no GPU. Appendix A lists the file behind each number.
- **SLM360:** six public pages on 360labs.ai, read on 28 October 2026 (Appendix A). These are the company's own claims. The model is *"available for licensing"* and is not public, so we could not run it on the same data or the same hardware.
- When two numbers come from different hardware or different tests, the text says so. Sizes are in decimal megabytes (1 MB = 1,000,000 bytes) unless MiB is written.

2 What the two systems are

	SLM360	**Kestrel**
Made by	360 Labs, a company	A research project (this repository)
Main job	Pick one intent (action) from a fixed list; 25–30 intents in its phone apps	Personal assistant: tasks, reminders, notes, writing, translation, questions about notes, small programs
How it decides	Four steps: patterns, an 85K-parameter model, Nano (6.4M parameters), Base (125M)	Three steps: keyword rules, a sentence encoder (multilingual-e5-small, about 118M parameters), Gemma-4-E2B
Writes text	Base (125M) is claimed to. In the phone apps: a tiny decoder (KAVACH) or SmolLM2-360M, a public model that is not SLM360 (SAGAR)	Yes, with Gemma-4-E2B (4-bit)
Languages	Paper: English only. SAGAR page: English and Hinglish commands	English and Hinglish, both measured
Runs on	Claims Linux, macOS, Windows, web browsers (WebAssembly), Android, iOS	One Linux laptop so far; the next version plans Linux, macOS and Windows
Open	Closed; sold under licence	Public models pinned by checksum; every number from a committed log

3 Where Kestrel is behind

Area	Kestrel	SLM360 (claimed)	Gap
Intent model file	132 MB (multilingual-e5-small, 8-bit)	4 MB (Nano, 4-bit); 0.85 MB in the KAVACH app	Kestrel 33× bigger (156× vs KAVACH)
Text model file	2.84 GB (Gemma-4-E2B, 4-bit)	63 MB (Base, 4-bit)	Kestrel 45× bigger
Both models vs the SAGAR app	2.97 GB	386 MB (Nano + SmolLM2-360M)	Kestrel 7.7× bigger
Memory, intent path	278 MiB peak (291 MB)	50 MB in total (paper)	Kestrel about 6× more
Memory with the text model	1,144 MiB the system cannot take back, plus the 2,702 MiB model file mapped	KAVACH needs a phone with at least 3 GB RAM for everything	Kestrel needs a large-memory device
Phones	None built, none measured	KAVACH, SAGAR and VAJRA are described as Android apps (KAVACH about 45 MB, VAJRA about 38 MB)	Kestrel has nothing here yet
Writing speed	12.1 tokens (word pieces) per second; an email took 8.5–10.8 s and a chat answer 23.2 s in one run	Base: *"<50ms/tok"*, that is over 20 tokens per second, on hardware not named	Kestrel about 1.7× slower per token, if the claim holds
Energy	Not measured (the energy counter needs root on this laptop)	0.001 Wh per query; method not given	No Kestrel number to compare
Users	One user; the pilot has not started	Apps described for defence clients (cannot be verified)	–

What this means. On everything that decides whether an assistant fits on a phone (file size, memory, phone builds), SLM360 is ahead by 6 to 45 times, and more against the KAVACH model. This gap is real, and it is the main thing Kestrel would have to fix to reach phones.

4 Where SLM360 is behind

Area	Kestrel	SLM360 (claimed)	Gap
Accuracy on CLINC150 (150 intents)	95.9% on 4,500 test sentences (intent step)[1]	85.1% (Nano page; number of test sentences not given)	Kestrel +10.8 points. SLM360's own reference model, BERT-base, scores 91.4%
Size of the test behind the headline	5,500 CLINC150 test items; the rules were fixed before any test item was read	98.0% on 49 SNIPS sentences; 100% on 50 sentences from a 10-intent part of Banking77 (the full set has 77)	Kestrel tested about 110× more items
Same tests at full size	–	Nano page: Banking77 with all 77 intents 88.3%; SNIPS 96.2%	SLM360's numbers disagree with each other (100% vs 88.3%)
Hinglish	90.0% of Hinglish requests (126/140) vs 92.9% of English (130/140) sent to the right skill	Paper: *"English only"*. SAGAR: two Hinglish examples, no numbers	Measured vs not measured
Writing	Emails, translation, rewriting; notes questions 19/19 answered right and 6/6 unanswerable ones declined; programs 21/24 right	KAVACH's sample: *"cobra team conducting clearance activity observed patrol grid 2ring north"*; its paper admits *"limited vocabulary and occasional grammatical imprecision"*. No sample from Base	SLM360's phone apps cannot do Kestrel's writing tasks
Requests that fit no action	1,000 such test items counted; Kestrel says it cannot do them	Paper: *"Fixed intent set"*	Not measured
Learning without forgetting	Measured: with EWC, forgetting 0.932, no better than without it (0.947); with replay −0.002	Model page: under 2% forgetting with EWC. Its own paper lists on-device learning as future work	SLM360 contradicts itself
Can others check it	Public models pinned by SHA-256; every number from a committed log	No public model; benchmark link returns 404; test phone never named	–

[1] The 95.9% counts the 4,500 test sentences that belong to one of the 150 intents, the same 150-intent setting as SLM360's number. On all 5,500 test items, including 1,000 that fit no intent, Kestrel's encoder alone scored 89.3% and its full cascade 87.4%.

SLM360 contradicts itself

- **Three sizes for one model:** 577K parameters and 848 KB (KAVACH paper); 6.4M parameters and 4 MB (Nano page); a 32 MB model and 50 MB in total (SLM360 paper).
- **"Pure Rust with zero external ML dependencies"**, but the same page lists an *"ONNX model (32MB)"*, and the KAVACH app, which uses SLM360, names *"TensorFlow Lite 2.14.0"* as its ML runtime.
- **"Proprietary" model:** the paper calls it proprietary, but its own benchmark commands load `gte-small-quantized.onnx`, a public embedding model.
- **Phones:** one section of the model page says *"iOS and Android planned"*; another lists *"Android via JNI bindings"*.
- **Speed vs Rasa:** *"4x faster"* in the abstract, *"3.8x faster"* in the results.

5 Not a contest: different jobs

- **Camera detection** of people and vehicles (YOLOv8n, 12 MB), **drone sounds**, **patrol routes** (A* path search) and **reports** (filled-in templates) are features of 360 Labs' apps, but they are not SLM360. VAJRA, the drone app, does not use SLM360 at all. Kestrel does not try to do any of these.
- **Why a 45 MB app can do so much:** in KAVACH, SLM360 is 0.85 MB. The rest is a standard 12 MB detector, ordinary code, a map and the screens. Picking one of 25 actions needs only a small model; writing free text needs a large one.
- **Fair pairing:** Kestrel's intent step does the same job as SLM360 Nano. Kestrel's Gemma step has no real counterpart in SLM360's phone apps.

6 Kestrel's own weak spots

Kestrel's logs also show where it falls short, regardless of SLM360:

- On CLINC150, the encoder alone (89.3%, 5.6 ms on average) beat Kestrel's full cascade (87.4%, 146 ms on average). Sending hard cases to Gemma made results worse there.
- On the small English–Hinglish set (252 items), the cascade (96.8%) stayed below always using Gemma (100%). The result is inconclusive.
- The English–Hinglish test sets are small and synthetic (252 and 280 items).
- Programs that read input: 3 of 6 right. Translation into Hinglish: 1 of 4 (development data only).
- Everything was measured on one laptop.

7 What this means if Kestrel goes to phones

- **Size is the main problem.** The intent step must shrink from 132 MB to a few MB. Gemma (2.84 GB) is the bigger question: either it stays on the laptop and the phone sends it only the hard requests, or it is measured on a phone with a lot of memory.
- **Going small has a cost.** If SLM360's numbers hold, a 4 MB model loses about 10 points on CLINC150 compared with Kestrel's encoder. A smaller Kestrel encoder must be measured, in English and in Hinglish, before it replaces the current one.

- **Nothing has been measured on a phone yet.** Every phone statement in this section is a plan, not a result.

8 Scorecard

Area	Ahead	By how much
Model file size	SLM360	33× (intent), 45× (text)
Memory, intent path	SLM360	about 6×
Phones	SLM360	Kestrel has no phone build
Energy figures	SLM360 (claim only)	Kestrel has no number
Writing speed per token	SLM360 (claim only)	about 1.7×
Intent accuracy, CLINC150	Kestrel	+10.8 points (95.9% vs 85.1%)
Hinglish	Kestrel	measured vs not measured
Writing, translation, notes questions	Kestrel	SLM360's phone apps do not do these
Size of the tests	Kestrel	5,500 items vs 49 and 50 sentences
Others can check it	Kestrel	public models and logs vs a closed model
In real use	SLM360 (claim only)	Kestrel: one user

A Sources

Kestrel (this repository)

- CLINC150 accuracy, speed and memory: `experiments/RESULTS.md`, Tables R1 and R2; per-tier accuracy on the 4,500 in-scope test sentences: `experiments/logs/E2/analysis_clinc150_test.json`.
- Forgetting (EWC, replay): `experiments/RESULTS.md`, Table R6.
- English and Hinglish decisions, notes questions, programs, translation into Hinglish, answer times, memory with Gemma loaded, Gemma speed: `paper/v3/generated/numbers.json`, keys `kt2.hg.en`, `kt2.hg.hi`, `qa.test.ans`, `qa.test.un`, `code.test.all`, `code.test.script`, `writer.final.translate-hinglish`, `acc.s.email`, `acc.s.emailhi`, `acc.s.chat`, `mmap.map.cold.anon`, `mmap.map.cold.file`, `e0.gemma.tg`.
- Model sizes: `models/MANIFEST.toml` (Gemma-4-E2B Q4_0: 2,841,481,184 bytes; the encoder's 32-bit source: 470,641,600 bytes, about 118M parameters) and the converted encoder file `models/multilingual-e5-small-q8_0.gguf` (132,440,544 bytes).
- Laptop: `docs/environment.md`. Service memory: `README.md`.

SLM360 (360labs.ai, read on 28 October 2026)

- Model page: `https://www.360labs.ai/models/slm360`
- Nano page: `https://www.360labs.ai/models/slm360-nano`
- SLM360 paper ("Technical Report, 2025", S. Kumar and P. R. Agrawal, not peer reviewed): `https://www.360labs.ai/research/slm360-paper`
- KAVACH paper ("Technical Report, 2026"): `https://www.360labs.ai/research/kavach-paper`
- VAJRA paper: `https://www.360labs.ai/research/vajra-paper`
- SAGAR page: `https://www.360labs.ai/product/sagar`
- The paper's raw-results link, `https://360labs.ai/lllm360/benchmarks`, returned 404.